1 Introduction

Imagine waking up on an ordinary Tuesday to find your company's entire global supply chain at a sudden standstill. No trucks are leaving the distribution centers. Warehouse shelves are empty of critical parts, while other storage areas overflow with inventory that will not be needed for months. No hackers stole data or wrote malicious code. Instead, a multi-million-dollar artificial intelligence system was misled by a fake weather report. Although this scenario may sound far-fetched, it is entirely possible. As organizations rely more on AI to run their operations, this risk demands serious attention. This report examines how such failures occur, why they matter, and what organizations can do to prevent them.

The integration of Artificial Intelligence (AI) into supply chain management is transforming traditional business operations through intelligent, autonomous, and data-driven capabilities. This evolution is embodied in the Artificial Intelligence Supply Chain Management Information System (ASCMIS), an advanced information system that combines large-scale data processing, AI-driven decision-making, and interconnected operational processes. These capabilities offer significant advantages in efficiency, automation, and decision-making, but they also introduce cybersecurity challenges that go beyond traditional information security concerns. The combination of sensitive data, autonomous decisions, and interactions with real-world operations creates a complex threat landscape. It calls for new approaches to identifying, assessing, and managing cybersecurity risks.

To address these challenges, this study adopts the EBIOS Risk Manager (EBIOS-RM) methodology. Developed by the French National Cybersecurity Agency (ANSSI), this threat-oriented risk analysis framework examines potential attackers, their motivations, and the attack paths they may exploit to compromise an information system. The methodology provides a structured foundation for developing the Information Systems Security Policy (ISSP) and the Business Continuity Plan (BCP), based on the risks identified and assessed throughout the analysis. The study also considers emerging cybersecurity requirements for AI-based systems, particularly those addressed by ISO/IEC 27090, which provides guidance on cybersecurity risks associated with artificial intelligence. The ASCMIS case study therefore offers an opportunity to examine the security implications of integrating AI into supply chain operations and to establish measures for protecting the system's information assets, ensuring operational resilience, and maintaining business continuity.